

# Factsheet: Beta distribution

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## Summary

A factsheet about the beta distribution.

Beta( $\alpha = 1.50$ ,  $\beta = 2.50$ )

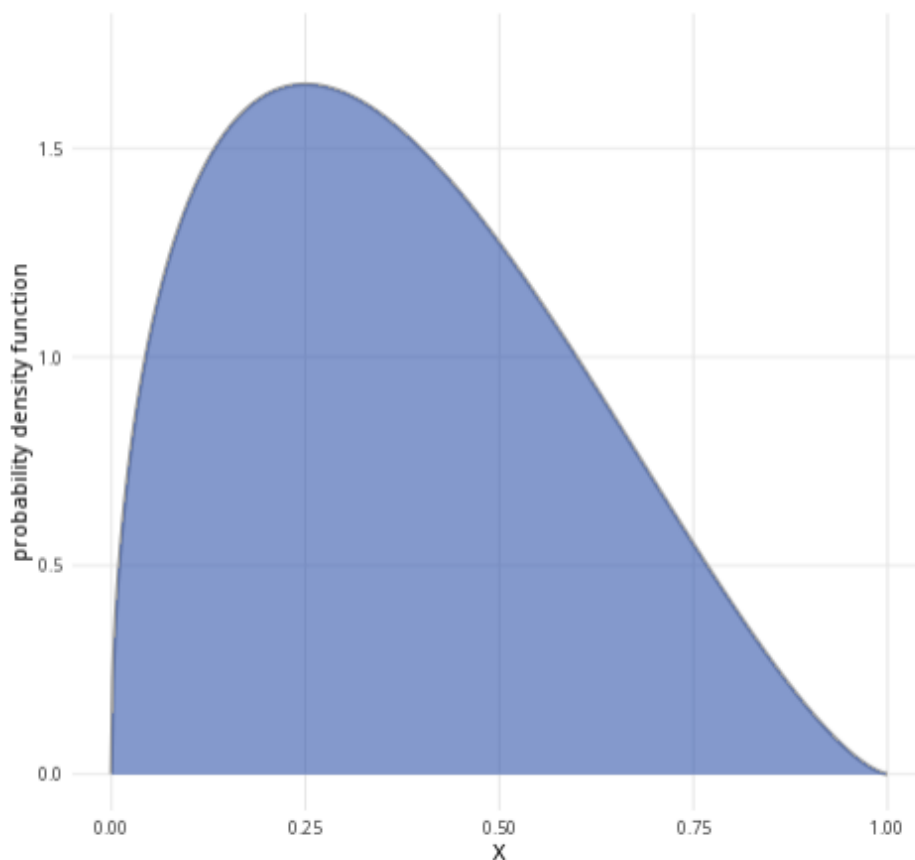


Figure 1: An example of the beta distribution with  $\alpha = 1.5$  and  $\beta = 2.5$ .

**Where to use:** The beta distribution is used to model the distribution of *probabilities* or proportions. Hence, the random variable  $0 \leq X \leq 1$ .

**Notation:**  $X \sim \text{Beta}(\alpha, \beta)$

**Parameters:** Two positive real numbers  $\alpha, \beta$ , which are shape parameters. These can be specified as follows in terms of  $n$  and  $k$  where  $n$  is the number of Bernoulli trials and  $k$  is the number of successes:

- $\alpha = k + 1$
- $\beta = n - k + 1$

| Quantity        | Value                                                                        | Notes                                                   |
|-----------------|------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>Mean</b>     | $\mathbb{E}(X) = \frac{\alpha}{\alpha + \beta}$                              |                                                         |
| <b>Variance</b> | $\mathbb{V}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$ |                                                         |
| <b>PDF</b>      | $\mathbb{P}(X = x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$   | $B(x, y)$ is the beta function                          |
| <b>CDF</b>      | $\mathbb{P}(X \leq x) = I_x(\alpha, \beta)$                                  | $I_x(a, b)$ is the regularized incomplete beta function |

**Example:** Cantor's Confectionery is visited by 10 customers, and 6 of them purchase something from the store. Taking the buying customers as successes and the total visiting customers as number of trials, there would be 6 successes, allowing you to find the following parameters:

- $\alpha = 6 + 1 = 7$
- $\beta = 10 - 6 + 1 = 5$

Then the distribution of the probabilities of a customer purchasing from Cantor's Confectionery can be expressed as  $X \sim \text{Beta}(7, 5)$ , meaning the first shape parameter is 7 and the second shape parameter is 5.

## Further reading

This interactive element appears in [Overview: Probability distributions](#). Please click this link to go to the guide.

## Version history

v1.0: initial version created 04/25 by tdhc and Michelle Arnetta as part of a University of St Andrews VIP project.

- v1.1: moved to factsheet form and populated with material from [Overview: Probability distributions](#) by tdhc.

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